Course	ECED 3202
Document Title	Lab 5: Multistage amplifiers
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Relevant Lectures & Examples:

The following material is relevant to this lab:

- Lecture 7 CD and CC
- Lecture 7 Examples
- All of Lecture 10
- Lecture 10 examples

Purpose/Scope

The purpose of this lab is to:

- Design a multi-stage amplifier that can simultaneously achieve a high input resistance, a low output resistance and a high gain
- Make appropriate use of CE and CC stages to meet requirements
- Use a CC voltage buffer to increase input resistance seen looking into a load

Common emitter amplifier

Problem statement

You are part of a startup company developing a wireless video projector. You oversee the team designing the amplifier for the audio subsystem and are asked to design a small power amplifier to drive an 8Ω *speaker located in the bottom of the device with at least 10mW of power with <5% THD and a minimum voltage gain of 15x. Your boss wants the design on his desk by Friday, so you blow the dust off your ECED3202 notes from undergrad and get to work designing…*

Architectural requirements

Requirement identifier	Requirement Description	Requirement marks
R5-01	The circuit shall be powered by a 5V supply	0
R5-02	The circuit shall drive an 8Ω speaker load coupled to the amplifier through a $100\mu F$ capacitor	0

Table 1: Architectural requirements

Performance requirements

Test conditions

- The operating frequency range shall be 500Hz to 5KHz
- Unless otherwise noted, requirements are specified with an 8Ω load connected to the output through a $100\mu F$ capacitor

Requirements

Requirement Identifier	Requirement Description	Requirement points
R5-03	The input impedance of the amplifier shall be $> 80k\Omega$ between 500Hz and 5KHz	5
R5-04	The amplifier shall produce at least 10mW of output power with a THD $<$ 5% at 1000Hz	5
R5-05	All transistors in the circuit shall operate within their safe operating range for current and power dissipation for inputs $\leq 100mVp$ (note that the response won't be linear with this amount of input signal, but that's ok)	5
R5-06	The gain of the amplifier shall be $>$ 15x (23.5dB) over the frequency range from 500Hz to 5KHz	5

Table 2: Performance requirements

Design Verification

Circuit Architecture

Architecture A3-01	
Requirement(s) being tested	R5-01, R5-02
Output files	Lab 5 Amplifer V1.asc, R5-01-circuit-diagram.png
Description of circuit	A 2N3906-based common emitter front end was combined with a 2N2222 Darlington pair to drive an 8 Ω speaker load, through a 100μF capacitor.
Acceptance criterion	Circuit contains all components defined in Table 1
Test result	PASS
Objective evidence	see A5-01-circuit-diagram.png

Performance Characteristics

Test T5-01	
Requirement(s) being tested	R5-03
Output files	T5-01-input-impedance.png
Description of simulation	Plotted a graph of input voltage by the input current for the frequencies between 500Hz and 5KHz.
Description of test output	The lowest value of the input impedance of the amplifier was over 100kΩ at the highest frequency
Acceptance criterion	The input impedance of the amplifier shall be >80kΩ between 500Hz and 5KHz
Test result	PASS
Objective evidence	see T5-01-input-impedance.png

Test T5-02	
Requirement(s) being tested	R5-04
Output files	T5-02-thd-and-power.png
Description of simulation	A 1-second transient analysis was performed on a 1 kHz, 30 mV signal. The FFT function was used to determine the THD. The power through the load was found by taking the average of V*I
Description of test output	The amplifier had 4.26% THD at 1000Hz while delivering 13.8mW
Acceptance criterion	The amplifier shall produce at least 10mW of output power with a THD < 5% at 1000Hz
Test result	PASS
Objective evidence	see T5-02-thd-and-power.png

Test T5-03	
Requirement(s) being tested	R5-05
Output files	T5-03-operating-power.png
Description of simulation	The peak current, voltage between collector and emitter, and the power dissipation were taken to be compared to their rated maximum values
Description of test output	The output of the measurements confirmed that the peak values were under the maximum ratings for all three transistor elements in the designed circuit
Acceptance criterion	All transistors in the circuit shall operate within their safe operating range for current and power dissipation for inputs $\leq 100mVp$
Test result	PASS
Objective evidence	See T5-03 Results table below

Test T5-03 Results						
Device	$I_{C,\mathbf{pk}}$	$I_{C,\mathbf{max}}$	$V_{CE,\mathbf{pk}}$	V_{CEO}	$P_{\mathbf{avg}}$	$P_{D,\mathbf{max}}$
Q1 (2N3906)	93 μA	200 mA	2.96 V	40 V	56 μW	625 mW
Q2 (2N2222)	1.9 mA	600 mA	2.89 V	40 V	0.87 mW	625 mW
Q3 (2N2222)	222 mA	600 mA	3.13 V	40 V	204 mW	625 mW

Test T5-04	
Requirement(s) being tested	R5-06
Output files	T5-04-gain.png
Description of simulation	The graph of output voltage over input voltage was taken in LTspice over 500Hz to 5kHz in decibels
Description of test output	The decibel readings for the gain of the output voltage over the input voltage is between 24dB and 25dB for the entirety of the frequency sweep
Acceptance criterion	The gain of the amplifier shall be >15x (23.5dB) over the frequency range from 500Hz to 5KHz
Test result	PASS
Objective evidence	see T5-04-gain.png